Take Home Assignment

Home Assessment – OCR Endpoint Challenge (/ocr)

1. Overview

Fullerton Health's engineering team processes thousands of medical documents every day. In this take-home assessment you will design and build a **single HTTP endpoint** that performs Optical Character Recognition (OCR), automatically detects the document type, and returns a structured JSON payload with the key data fields we need for downstream claim adjudication.

The exercise is intended to assess your skills in:

- Document classification & information extraction (classical / ML / deep-learning—your choice) API & micro-service design
- Clean, maintainable code and automated tests
- Written communication (README/comments)

2. Functional Requirements

2.1 Endpoint

Method	Path	Description
POST	/ocr	Accepts a document image/PDF and returns extraction results

2.2 Supported document types

Internal code	Human label
referral_letter	Referral Letters
medical_certificate	Medical Certificates

receipt	Receipts

The service **must** correctly identify which of the three types was supplied. If the document type is outside this list, return HTTP 422 with a message "unsupported_document_type" .

2.3 Field extraction

For each recognised type, extract the fields below **exactly as named**. Unless otherwise noted, missing or un-parsable values should be returned as null .

2.3.1 Referral Letter

Key	Description
claimant_name	Patient Name
provider_name	Provider / Lab name (must **not** contain the literal string "Fullerton Health")
signature_presence	true if a handwritten signature is detected, else false
total_amount_paid	Integer with all currency symbols, separators and decimals removed
total_approved_amount	Same rules as above
total_requested_amount	Same rules as above

2.3.2 Medical Certificate

Key	Description
claimant_name	Claimant Name
claimant_address	Address
claimant_date_of_birth	DD/MM/YYYY
diagnosis_name	Diagnosis

discharge_date_time	DD/MM/YYYY
icd_code	ICD code
provider_name	Provider / Lab name (no "Fullerton Health")
submission_date_time	Admission datetime (DD/MM/YYYY)
date_of_mc	Date of MC (DD/MM/YYYY)
mc_days	Integer number of MC days

2.3.3 Receipt

Key	Description
claimant_name	Claimant Name
claimant_address	Address
claimant_date_of_birth	DD/MM/YYYY
provider_name	Provider / Lab name (no "Fullerton Health")
tax_amount	Integer, stripping all separators & decimals
total_amount	Integer, stripping all separators & decimals

3. API Contract

3.1 Request

```
POST /ocr
Content-Type: multipart/form-data
file=<binary document>
```

Accept both PDF and common image formats (JPG/PNG).

3.2 Successful Response (HTTP 200)

```json
{
  "message": "Processing completed.",
  "result": {
    "document_type": "referral_letter",
    "total_time": 3.04,
    "finalJson": {
      "claimant_name": "…",
      "provider_name": "…",
      "signature_presence": true,
      "total_amount_paid": 3000000,
      "total_approved_amount": 3000000,
      "total_requested_amount": 3000000
    }
  }
}
```

`*_elapsed` * timings are optional but nice to have.*

3.3 Error Responses

Status	Condition	Example body
400	No file / invalid MIME type	`{ "error": "file_missing" }`
422	Unsupported document type	`{ "error": "unsupported_document_type" }`
500	Unhandled exception	`{ "error": "internal_server_error" }`

4. Deliverables

- **Source code** (public Git repository or zip)
- **README.md** with:
  - Setup & run instructions
  - How to extend to new document types
  - Sample curl commands and (optional) Postman collection

- **Assignment summary report** in PDF format, including
  - API input and outputs result on assessment documents (preferably in json format)
  - **Pipeline Architecture Diagram** including your architecture design and data flow details

Tip: You do **not** need to train a full model; simple pattern rules or off-the-shelf LLM calls are acceptable as long as they meet the requirements.

5. Appendix

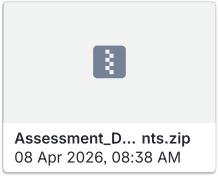
5.1 Suggested Tools & Libraries

- OCR: **Tesseract** (pytesseract), **PaddleOCR**, **Google Vision API**
- PDF handling: pdf2image, PyMuPDF, pdfplumber
- Web framework: **FastAPI**, **Flask**, or any framework you're proficient in
- ML / Prompting: scikit-learn, transformers, **OpenAI GPT** or other LLM API (if you have API credits), Open Source LLMs (if you have enough compute resource)

5.2 Contact

For questions, please email jeff.lu@fullertonhealth.com. We typically reply within one business day.

5.3 Sample Document File

Assessment_Documents.zip folder

- medical_certificate.pdf
- receipt.pdf
- referral_letter.pdf